

Answer Key

Q1	(3)	Q36	(2)
Q2	(2)	Q37	(3)
Q3	(2)	Q38	(3)
Q4	(4)	Q39	(1)
Q5	(3)	Q40	(1)
Q6	(2)	Q41	(2)
Q7	(3)	Q42	(3)
Q8	(4)	Q43	(1)
Q9	(1)	Q44	(2)
Q10	(1)	Q45	(3)
Q11	(4)	Q46	(3)
Q12	(1)	Q47	(3)
Q13	(1)	Q48	(2)
Q14	(1)	Q49	(1)
Q15	(4)	Q50	(1)
Q16	(4)	Q51	(3)
Q17	(3)	Q52	(1)
Q18	(2)	Q53	(3)
Q19	(1)	Q54	(1)
Q20	(1)	Q55	(4)
Q21	(1)	Q56	(3)
Q22	(4)	Q57	(2)
Q23	(2)	Q58	(3)
Q24	(3)	Q59	(3)
Q25	(2)	Q60	(1)
Q26	(3)	Q61	(1)
Q27	(2)	Q62	(3)
Q28	(4)	Q63	(2)
Q29	(4)	Q64	(3)
Q30	(3)	Q65	(4)
Q31	(1)	Q66	(2)
Q32	(1)	Q67	(4)
Q33	(3)	Q68	(4)
Q34	(1)	Q69	(3)
Q35	(3)	Q70	(3)



Q71 (3)
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Q197 (4)
Q198 (3)
Q199 (1)
Q200 (1)



Hints & Solutions

Q1 Text Solution:

(3)

$$Y = \vec{A} + \vec{B} = \vec{A} \cdot \vec{B} = A \cdot B$$

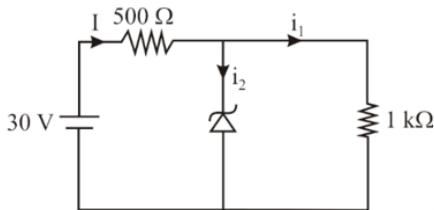
[New NCERT Class 12th, Page No. 480]

Q2 Text Solution:

(2)

Potential drop across dropping resistance = 20 V

$$\text{Current in dropping resistance } i = \frac{20}{500} = 40 \text{ mA}$$



$$\text{Current in } 1\text{k}\Omega, i_1 = \frac{10}{1000} = 10 \text{ mA}$$

$$\text{Current in Zener diode} = (40 - 10)\text{mA} = 30 \text{ mA}$$

[New NCERT Class 12th, Page No. 472]

Q3 Text Solution:

(2)

$$\frac{3h}{2\pi} = n \left(\frac{h}{2\pi} \right) \quad (\because n=3)$$

$$K_n = \frac{K_1}{(3)^2} = \frac{13.6}{9} = 1.51 \text{ eV}$$

Q4 Text Solution:

(4)

Only +ve part passes out.

[New NCERT Class 12th, Page No. 484]

Q5 Text Solution:

(3)

$$I = \frac{I_0}{2} \cos^2 60^\circ = \frac{I_0}{8}$$

[New NCERT Class 12th, Page No. 377]

Q6 Text Solution:

(2)

$$x_n = \frac{nD\lambda}{d} \Rightarrow \frac{d}{2} = \frac{nD\lambda}{d}, n = \frac{d^2}{2\lambda D}$$

[New NCERT Class 12th, Page No. 365]

Q7 Text Solution:

(3)

$$\text{Highest order of minima} = \frac{d}{\lambda} = \frac{3\lambda}{\lambda} = 3$$

$$\text{Total number of minima} = 2 \times 3 = 6$$

[New NCERT Class 12th, Page No. 365]

Q8 Text Solution:

(4)

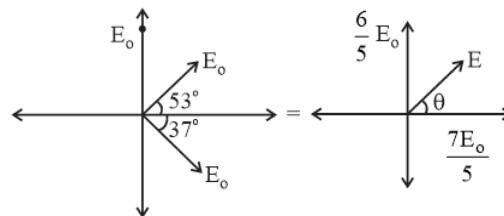
Stopping potential of B is greater than A, hence incident frequency of B is greater than A by equation

$$h\nu = \phi_0 + eV_0$$

[New NCERT Class 12th, Page No. 428]

Q9 Text Solution:

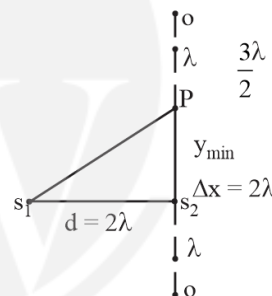
(1)



$$E(t) = \frac{\sqrt{85}}{5} \sin\left(\omega t + \tan^{-1} \frac{6}{7}\right)$$

Q10 Text Solution:

(1)



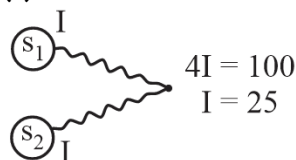
$$s_1P - s_2P = \frac{3\lambda}{2}$$

$$\sqrt{(2\lambda)^2 + y^2} - y = \frac{3\lambda}{2}$$

$$y_{\min} = \frac{7\lambda}{12}$$

Q11 Text Solution:

(4)



Now

$$I_{\max} = \left(\sqrt{25} + \sqrt{\frac{64}{100} \times 25} \right)^2 = \left(5 + \frac{8}{10} \times 5 \right)^2 = 81$$

Q12 Text Solution:

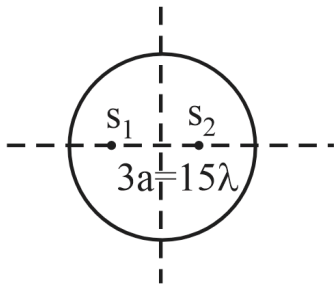
(1)



Android App

iOS App

PW Website



$$3a = 15\lambda$$

Total maxima

$$= 60$$

Q13 Text Solution:

(1)

$$\Delta x = \left(2n - 1\right) \frac{\lambda}{2} = 1, 2, 3, 4, \dots \text{ and } \Delta x = n\lambda$$

Q14 Text Solution:

(1)

$$\beta^1 = \frac{\beta}{\mu} = \frac{0.4}{4/3} \beta^1 = 0.30 \text{ mm}$$

Q15 Text Solution:

(4)

$$y = \frac{n\lambda D}{d} = 3 \text{ cm} \Rightarrow y^1 = \frac{n\lambda' D}{d} = \frac{n\lambda D}{\mu d} = 2 \text{ cm}$$

Q16 Text Solution:

(4)

$$B_n = \frac{\mu_0 I_n}{2r_n} \text{ or } B_n \propto \frac{I_n}{r_n} \propto \frac{(f_n)}{r_n}$$

$$\therefore B_n \propto \frac{(v_n/r_n)}{r_n} \propto \frac{v_n}{(r_n)^2}$$

$$\propto \frac{(Z/n)}{(n^2/Z)^2} \propto \frac{Z^3}{n^5}$$

Q17 Text Solution:

(3)

$$F = eB = \frac{mv}{R}$$

$$v = \frac{eBR}{m} - 1$$

$$k_{\max} = \frac{1}{2}mv^2 = \frac{hc}{\lambda} - \phi$$

$$\Rightarrow \phi = \frac{hc}{\lambda} - \frac{mv^2}{2}$$

$$\frac{12400}{2475} \text{ eV} - m \times \frac{e^2 B^2 R^2}{2m^2}$$

$$= 5 \text{ eV} - \frac{|7 \times 10^{11} \times 10^{-10}|}{2 \times 17} \text{ eV}$$

$$= (5 - 0.5)$$

Q18 Text Solution:

(2)

$$\frac{hc}{\lambda} = \frac{hc}{\lambda_0} + eV$$

$$\frac{hc}{2\lambda} = \frac{hc}{\lambda_0} + \frac{eV}{4}$$

$$\frac{hc}{\lambda} - \frac{hc}{\lambda_0} = eV$$

$$\frac{hc}{2\lambda} - \frac{hc}{\lambda_0} = \frac{eV}{4}$$

$$\frac{\text{iii}}{\text{iv}}$$

$$\frac{4}{2\lambda} - \frac{4}{\lambda_0} = \frac{1}{\lambda} - \frac{1}{\lambda_0}$$

Q19 Text Solution:

(1)

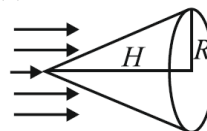
$$f_c > f_a = f_b$$

and

$$I_c = I_b > I_a$$

Q20 Text Solution:

(1)



$$F = \frac{F}{A} = \frac{I}{C} \Rightarrow F = \frac{IA'}{C}$$

$$F = \frac{I\pi R^2}{c}$$

Q21 Text Solution:

(1)

$$F = \frac{0.3P}{c} = \frac{2(0.7)P}{c} = \frac{0.3P}{c} + \frac{1.4P}{c}$$

$$= \frac{1.7P}{c}$$

Q22 Text Solution:

(4)



$$T \sin \theta = \frac{2IA}{c}$$

$$T \cos \theta = mg$$

$$\tan \theta = \frac{2IA}{Mgc}$$

$$\theta = \tan^{-1} \frac{2IA}{Mgc}$$

Q23 Text Solution:

(2)

$$F = \frac{2IA}{C}, F_{sp} = 3kx$$

$$W_{all} = DK$$

$$W_{ft} + W_{sp} = 0; \quad x_m = 2F/3K$$

Q24 Text Solution:

(3)

$$\Delta P = 2P \cos \theta; \quad F = \frac{dp}{dt}; \quad n = \frac{p}{h\nu};$$

$$F = \frac{2p \cos \theta}{c}$$

Q25 Text Solution:

(2)

$$\frac{P}{c} = mgP = mcg = 3 \times 10^7 \text{ w}$$

Q26 Text Solution:

(3)

$$F_R = \frac{(1+\alpha)IA}{c} = \frac{1.3\pi R^2}{c}$$

Q27 Text Solution:

(2)

The combined mass of the 50 hydrogen atoms and 70 neutrons

$$\begin{aligned} &= (20 \times 1.007825 + 70 \times 1.008665) \text{ amu} \\ &= 120.99780 \text{ amu} \end{aligned}$$

Therefore the mass defect is

$$\begin{aligned} \Delta m &= (120.99780 - 119.902199) \text{ amu} \\ &\approx 1.096 \text{ amu} \end{aligned}$$

The binding energy of the nucleus can now be obtained using equation

$$\begin{aligned} E_b &= \Delta m \cdot c^2 \\ &= (1.096 \text{ amu})(931 \text{ MeV/amu}) \\ &= 1020.38 \text{ MeV} \end{aligned}$$

Q28 Text Solution:

(4)

Mass defect,

$$\begin{aligned} \Delta m &= 4.00389 - (2 \times 1.00893 + 2 \times 1.00813) \\ &= -0.03023 \text{ amu} \end{aligned}$$

Binding energy,

$$\begin{aligned} \Delta E &= \Delta m \times 931.5 \text{ MeV} \\ &= 0.03023 \times 931.5 = 28.1 \text{ MeV} \end{aligned}$$

Q29 Text Solution:

(4)

$$Q = -200 \times 7.4 + 110 \times 8.2 + 90 \times 8.2 = 160 \text{ MeV}$$

Q = diff. of total energy of product and reactant.

Q30 Text Solution:

(3)

$$\begin{aligned} \text{Energy released} &= E_{\text{Product}} - E_{\text{Reactant}} \\ &= c - (a + b) \end{aligned}$$

Q31 Text Solution:

(1)

$$\frac{1}{v_1} + \frac{1}{30} = \frac{1}{10}$$

$$\Rightarrow v_1 = 15 \text{ cm}$$

$$\Rightarrow \frac{1}{v_2} - \frac{1}{10} = -\frac{1}{10}$$

$$\Rightarrow v_2 = \infty \Rightarrow v_3 = 30 \text{ cm}$$

$$OV_3 = 75 \text{ cm}$$

Q32 Text Solution:

(1)

$$v_i = v_0$$

$$v_i = 2 \text{ cm/sec}$$

Q33 Text Solution:

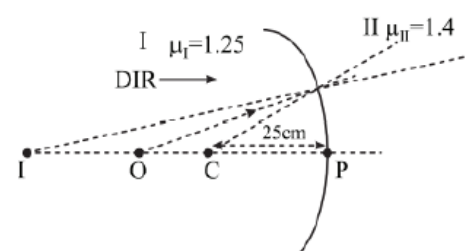
(3)

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

$$\Rightarrow \frac{1.4}{v} - \frac{1.25}{-40} = \frac{1.4 - 1.25}{-25}$$

$$\Rightarrow -\frac{1.4}{v} = \frac{0.15}{25} + \frac{1.25}{40} = \frac{7.45}{200}$$

$$\Rightarrow v = -\frac{200 \times 1.4}{7.45} = -37.58 \text{ cm}$$



Q34 Text Solution:

(1)



By snell law

$$\mu \sin \theta = \text{constant.}$$

Q35 Text Solution:

(3)

$$P_{\text{eff}} = 2P = \frac{2(\mu - 1) \times 2}{10} \mu = \frac{9}{8}$$

Q36 Text Solution:

(2)

$$n_2 > n$$

convex lens behave like concave lens.

Q37 Text Solution:

(3)

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

Q38 Text Solution:

(3)

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

Q39 Text Solution:

(1)

$$\mu = \frac{\sin\left(\frac{A+\delta}{2}\right)}{\sin\frac{A}{2}}$$

Q40 Text Solution:

(1)

$$\delta + A = i + e$$

Q41 Text Solution:

(2)

As x-axis is the normal to the mirror. so, only x component will change and y,z component will remain same. in x directions $V_i = 2V_m - V_o$. $V_i = 0$. in x direction. so $V_i = -3\hat{i} + 4\hat{k}$.

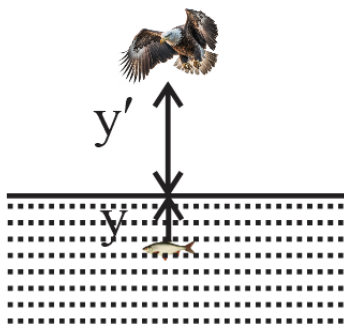
Q42 Text Solution:

(3)

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

Q43 Text Solution:

(1)



Effective height of the bird as seen by the fish,

$$Y = y + \mu y'$$

Differentiating both sides w.r.t t

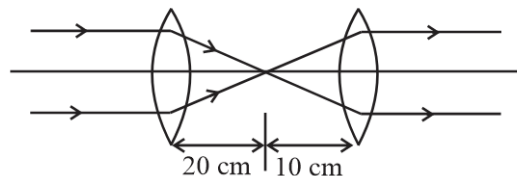
$$\frac{dY}{dt} = \frac{dy}{dt} + \mu \frac{dy'}{dt}$$

From question,

$$\frac{dY}{dt} = 9 \text{ ms}^{-1}; \frac{dy}{dt} = 3 \text{ ms}^{-1} \therefore 9 = 3 + \frac{4}{3} \frac{dy'}{dt} \Rightarrow \frac{dy'}{dt} = \frac{6 \times 3}{4} = 4.5 \text{ ms}^{-1}$$

Q44 Text Solution:

(2)



Q45 Text Solution:

(3)

Path difference

$$= (\mu_2 - \mu_1)t = 12480 \text{ \AA}$$

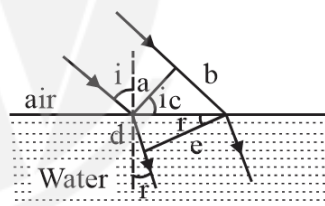
For maxima,

$$n\lambda = 12480 \text{ \AA} \lambda_1 = 12480 \text{ \AA} \lambda_2 = 6240 \text{ \AA} \lambda_3 = 4160 \text{ \AA} \lambda_4 = 3120 \text{ \AA}$$

Therefore, only 6240 \text{ \AA} and 4160 \text{ \AA} exist in the spectrum.

Q46 Text Solution:

(3)



From snell's law,

$$\mu_a \sin i = \mu_w \sin r$$

$$\frac{b}{c} = \mu_{wa} \frac{d}{c}$$

$$\Rightarrow \mu_{wa} = \frac{b}{d}$$

Q47 Text Solution:

(3)

Relation of intensity with phase difference is

$$I = I_0 \cos^2\left(\frac{\Delta\phi}{2}\right)$$

Here I_0 is intensity of central bright fringe

On substitution,



$$\Rightarrow \frac{I}{I_0} = \cos^2 \left(\frac{\frac{\pi}{4}}{2} \right) = \cos^2 \left(\frac{\pi}{8} \right)$$

$$\Rightarrow \frac{I}{I_0} = 0.853$$

Q48 Text Solution:

(2)

Given, $\Delta = \beta$

$$\text{or } \frac{D(\mu-1)t}{d} = \frac{D\lambda}{d}$$

$$\therefore t = \frac{\lambda}{(\mu-1)}$$

Q49 Text Solution:

(1)

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi$$

Q50 Text Solution:

(1)

$$(E_k)_{\max} = eV_s V_s = \frac{3ev}{e} = 3 \text{ V}$$

Q51 Text Solution:

(3)

The limiting line of Balmer series refers to the transition of electron from ∞ to 2^{nd} orbit.

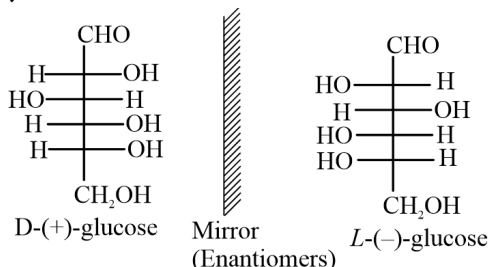
$$\nu = c \cdot \bar{\nu} = 3 \times 10^{10} \times 109677 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \text{ s}^{-1}$$

$$= 3.29 \times 10^{15} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \text{ s}^{-1}$$

$$= 8.22 \times 10^{14} \text{ s}^{-1} \left(\because n_1 = 2, n_2 = \infty \right)$$

Q52 Text Solution:

(1)

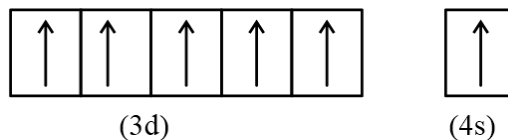


Q53 Text Solution:

(3)

From $[Ar] 3d^4 4s^2$, if one electron shifts from $4s$ orbital to vacant $3d$ -orbital, electronic

configuration is more symmetrical which is $[Ar]$



This leads to extra stability, hence Cr_{24} has electronic configuration is $[Ar] 3d^5 4s^1$.

Q54 Text Solution:

(1)

Calculated Vapour density,

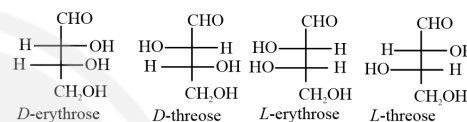
$$D = \frac{\text{Mol. wt.}}{2} = \frac{92}{2} = 46$$

Observed vapour density, $d = 30$

$$\alpha = \frac{D-d}{(n-1)d} = \frac{46-30}{30} = \frac{16}{30} = 0.533 = 53.3\%$$

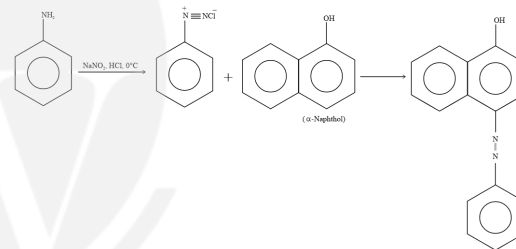
Q55 Text Solution:

(4)



Q56 Text Solution:

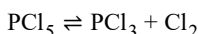
(3)



Q57 Text Solution:

(2)

According to Le-Chatelier's principle, the increase in pressure shift the equation in the direction of lesser number of gaseous moles.



$\therefore$ Some PCl_3 and Cl_2 will give more of PCl_5

Q58 Text Solution:

(3)

$$K.E. = \frac{1}{2}mv^2$$

$$v = \sqrt{\frac{2K.E.}{m}} = \sqrt{\frac{2 \times 6 \times 10^{-25}}{9.1 \times 10^{-31}}}$$

$$= 1.148 \times 10^3 = 1148 \text{ m s}^{-1}$$

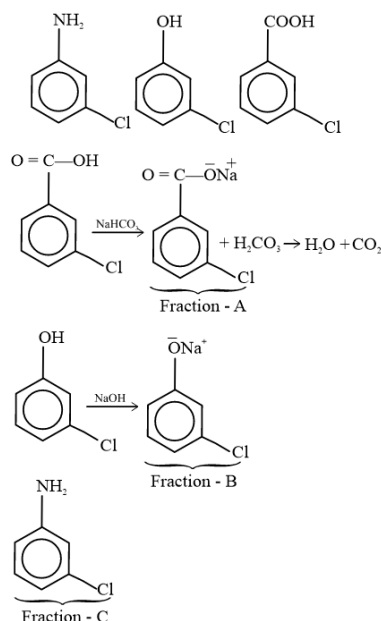
According to de-Broglie equation,

$$\lambda = \frac{h}{mv} = \frac{6.626 \times 10^{-34}}{9.1 \times 10^{-31} \times 1148} = 6342 \text{ \AA}$$

Q59 Text Solution:

(3)



**Q60 Text Solution:**

(1)

It is an example of Hoffmann bromamide reaction in which the amine formed has one carbon less than that of the parent amide.

Q61 Text Solution:

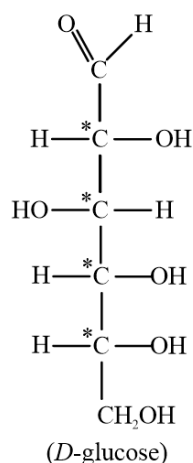
(1)

Kjeldahl's method for estimation of nitrogen is not applicable for nitrobenzene $C_6H_5NO_2$ because reaction with H_2SO_4 , nitrobenzene cannot give ammonia.

Q62 Text Solution:

(3)

Glucose contains four chiral carbon

**Q63 Text Solution:**

(2)

$$\text{Moles of water gas } (CO + H_2) = \frac{60}{30} = 2$$

$$\begin{aligned} \text{Moles of CO} &= \text{Moles of } H_2 = 1; \text{ Moles of steam} \\ &= \frac{18}{18} = 1 \end{aligned}$$

	CO	+	H ₂ O	⇌	CO ₂	+	H ₂
Initial conc.	1		1		0		1
At equilibrium conc.	1 - x		1 - x		x		1 + x

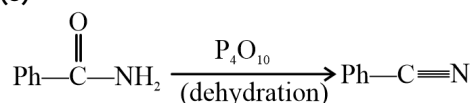
$$K_a = 3 = \frac{[CO_2][H_2]}{[CO][H_2O]} = \frac{(x)(1+x)}{(1-x)(1-x)}$$

$$x = 0.5 \text{ mol of } CO_2 = 0.5$$

Hence, the mass of $CO_2 = 22 \text{ g}$

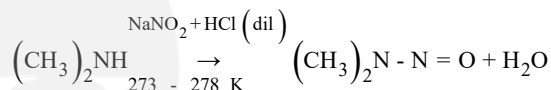
Q64 Text Solution:

(3)

**Q65 Text Solution:**

(4)

Secondary amines, both aliphatic and aromatic, react with nitrous acid to give yellow oily N-nitroso amines.

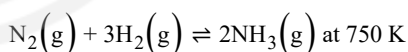
**Q66 Text Solution:**

(2)

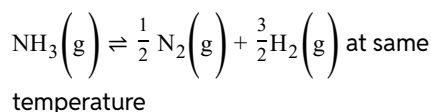
Anomer means configuration at α -carbon

Q67 Text Solution:

(4)



$$K = \frac{[NH_3]^2}{[N_2][H_2]^3} = 49$$



$$K' = \frac{[N_2]^{1/2}[H_2]^{3/2}}{[NH_3]}, K' = \frac{1}{\sqrt{49}} = \frac{1}{7}$$

Q68 Text Solution:

(4)

NH_4^+ (Ammonium ion)

- Nessler's Reagent (alkaline solution of potassium tetraiodomercurate, $K_2[HgI_4]$) is commonly used to detect the presence of

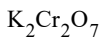


ammonia (NH_3) or ammonium ions (NH_4^+) in a solution.

- When ammonia is present, Nessler's reagent produces a characteristic yellow to brown precipitate or coloration due to the formation of mercury-ammonia complexes.

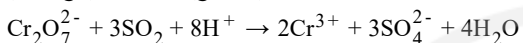
Q69 Text Solution:

(3)



SO_2 reacts with $\text{K}_2\text{Cr}_2\text{O}_7$:

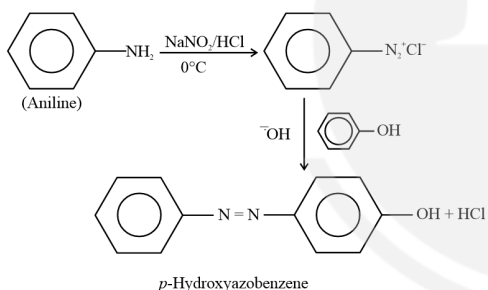
When SO_2 gas is passed through an acidified solution of potassium dichromate ($\text{K}_2\text{Cr}_2\text{O}_7$), the solution changes color from orange to green. This happens because SO_2 reduces $\text{Cr}_2\text{O}_7^{2-}$ (orange) to Cr^{3+} (green).



CO_2 : Carbon dioxide does not react with $\text{K}_2\text{Cr}_2\text{O}_7$ under these conditions, so no color change is observed.

Q70 Text Solution:

(3)



Q71 Text Solution:

(3)

Equilibrium constant is independent of initial concentrations of the reactants and products.

Q72 Text Solution:

(3)

Benedict's solution is more stable than Fehling's solution and is not affected by substances like creatine and uric acid present in urine. Hence, it is preferred to detect glucose in urine.

Q73 Text Solution:

(2)

$$\Delta G^\circ = -2.303RT \log K_{(sp)}$$

$$63.3 \times 10^3 \text{ J} = -2.303 \times 8.314 \times 331 \times \log K_{(sp)}$$

$$63.3 \times 10^3 \text{ J} = -6337.70 \times \log K_{(sp)}$$

$$\log K_{(sp)} = -\frac{63.3 \times 10^3}{6337.70} = -9.98 \approx -10$$

$$K_{(sp)} = \text{antilog}(-10) = 10^{-10}$$

Q74 Text Solution:

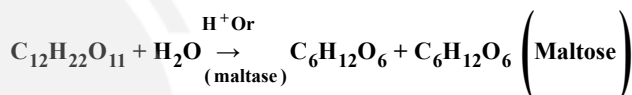
(2)

Ferric Sulphate reacts with $\text{K}_4[\text{Fe}(\text{CN})_6]$

- Prussian Blue is a complex compound with the formula $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3 \cdot x\text{H}_2\text{O}$.
- It is formed when ferric ions (Fe^{3+}) from ferric sulfate or another ferric salt react with ferrocyanide ions ($\text{Fe}(\text{CN})_6^{4-}$) from potassium ferrocyanide.
- The reaction is as follows:
- $4\text{Fe}^{3+} + 3[\text{Fe}(\text{CN})_6]^{4-} \rightarrow \text{Fe}_4[\text{Fe}(\text{CN})_6]_3$ (Prussian Blue)

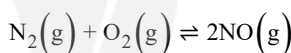
Q75 Text Solution:

(1)



Q76 Text Solution:

(2)



For this reaction $\Delta n_g = 0$

$$\text{Since } K_p = K_c (\text{RT})^{\Delta n_g}$$

$$\therefore K_p = K_c = 4.0 \times 10^{-6}$$

Q77 Text Solution:

(4)



- When ammonium hydroxide (NH_4OH) reacts with mercurous chloride (Hg_2Cl_2), a black precipitate of metallic mercury (Hg) is formed along with mercuric ammonium chloride.
- The reaction is:
- $\text{Hg}_2\text{Cl}_2 + 2\text{NH}_4\text{OH} \rightarrow \text{Hg} + \text{HgNH}_2\text{Cl} + 2\text{H}_2\text{O}$

Ag_2S : This is already a black precipitate and does not react with NH_4OH .

AgCl : Reacts with NH_4OH to form a soluble

complex $[\text{Ag}(\text{NH}_3)_2]^+$, so no black precipitate is observed.



HgCl_2 : Reacts with NH_4OH to form a white precipitate.

Q78 Text Solution:

(1)

$$w = 0.189 \text{ g}$$

$$\text{wt. of AgCl} = 0.287 \text{ g}$$

$$\% \text{ of Cl} = \frac{35.5 \times 0.287 \times 100}{143.5 \times 0.189} = 37.57 \%$$

Q79 Text Solution:

(4)

graph (1) represents 1s-orbital

graph (2) represents 3s-orbital

graph (3) incorrect.

graph (4) represents the correct radial distribution curve for 3p-orbital.

Q80 Text Solution:

(2)

In non-aqueous solvents and in vapour phase, basic strength of alkyl amines follows the order :
Tertiary amines > secondary amines > primary amines

Q81 Text Solution:

(4)

In a multi-electron atom, orbitals (iv)

$n = 3, l = 2, m = 1$ and (v) $n = 3, l = 2, m = 0$ will have the same energy, as energy corresponds to $(n + l)$ value.

Q82 Text Solution:

(2)

$$\text{Degree of dissociation, } \alpha = \frac{3.7}{100} = 0.037$$

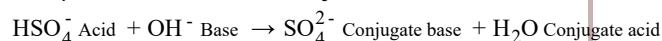
According to Ostwald's formula,

$$K_a = \alpha^2 C = (0.037)^2 \times 0.10 = 1.369 \times 10^{-4} \approx 1.4 \times 10^{-4}$$

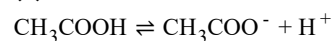
Q83 Text Solution:

(3)

Proton donors are Brönsted-Lowry acid. Proton acceptors are Brönsted-Lowry base.

**Q84 Text Solution:**

(4)



$$K_{\text{ion}} = \frac{[\text{CH}_3\text{COO}^-][\text{H}^+]}{[\text{CH}_3\text{COOH}]}$$

$$[\text{CH}_3\text{COOH}] = \frac{(3.4 \times 10^{-4}) \times (3.4 \times 10^{-4})}{1.7 \times 10^{-5}} = 6.8 \times 10^{-3} \text{ M}$$

Q85 Text Solution:

(4)

A mixture of 1° , 2° and 3° -amines can be distinguished by Hinsberg's method or Hofmann's method. In Hinsberg's method benzene sulphonyl chloride $\text{C}_6\text{H}_5\text{SO}_2\text{Cl}$ (Hinsberg's reagent) is used. 1° and 2° -amines react to give sulphonamide derivatives with loss of HCl , whereas 3° -amines do not give any isolable products other than the starting amine. In the latter case a quaternary "onium" salt may be formed as an intermediate, but this rapidly breaks down in water to liberate the original 3° -amine.

Q86 Text Solution:

(1)

$$K.E. = \frac{1}{2}mv^2, \text{ Potential energy} = -\frac{Ze^2}{r}$$

We know that in a stable atom, electrostatic force = centrifugal force i.e., $\frac{Ze^2}{r^2} = \frac{mv^2}{r} \Rightarrow P.E. = -mv^2$

Total energy = K.E. + P.E.

$$= \frac{1}{2}mv^2 + (-mv^2) = -\frac{1}{2}mv^2$$

$$\Rightarrow \frac{K.E.}{\text{Total energy}} = \frac{\frac{1}{2}mv^2}{-\frac{1}{2}mv^2} = -1$$

Q87 Text Solution:

(4)

$$\text{For weak bases: } [\text{OH}^-] = \sqrt{K_b \times C}$$

$$\text{pH} = 9.7 \text{ Thus, } \text{pOH} = 14 - 9.7 = 4.3$$

$$-\log[\text{OH}^-] = 4.3$$

$$[\text{OH}^-] = 5 \times 10^{-5}$$

$$5 \times 10^{-5} = \sqrt{K_b \times 0.004} \Rightarrow K_b \times 0.004 = 25 \times 10^{-10}$$

$$K_b = \frac{25}{4 \times 10^{-3}} \times 10^{-10} = 6.25 \times 10^{-7}$$

Q88 Text Solution:

(3)

The uncertainty in the speed (Δv) is 2%, i.e.,

$$45 \times \frac{2}{100} = 0.9 \text{ m s}^{-1}$$

$$\Delta x = \frac{h}{4\pi m \Delta v}$$

$$= \frac{6.626 \times 10^{-34} \text{ J s}}{4 \times 3.14 \times 40 \times 10^{-3} \text{ kg} \times 0.9 \text{ m s}^{-1}} = 1.46 \times 10^{-33} \text{ m}$$

Q89 Text Solution:

(3)

Glycine $\text{H}_3\text{N}^+\text{CH}_2\text{COO}^-$ is more acidic. Instead of K_b value, the second given K_a value is



corresponding to the K_b value

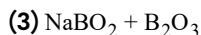
($\because K_a \times K_b = 10^{-14}$). Hence, overall ionization constant,

$$K = K_{a_1} \times K_{a_2} = 4.5 \times 10^{-3} \times 1.7 \times 10^{-10} = 7.65 \times 10^{-13}$$

$$H^+ = \sqrt{KC} = \sqrt{7.65 \times 10^{-13} \times 0.01} = \sqrt{0.765 \times 10^{-14}} \\ = 0.87 \times 10^{-7} M$$

$$pH = -\log(0.87 \times 10^{-7}) = 7.06$$

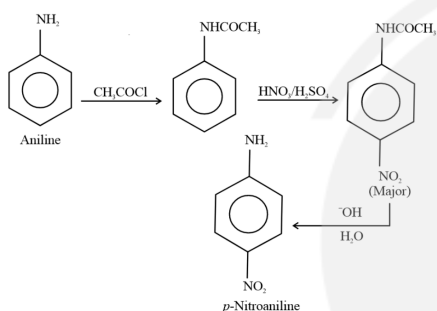
Q90 Text Solution:



This corresponds to sodium metaborate combined with boric oxide, commonly formed during bead tests in analytical chemistry.

Q91 Text Solution:

(3)



Q92 Text Solution:

(4)

$$pH = 7 + \frac{1}{2}(pK_a - pK_b) = 7 + \frac{1}{2}(3.75 - 4.75) = 6.5$$

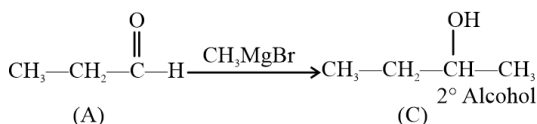
Q93 Text Solution:

(1)

Adenine forms hydrogen bonds with thymine whereas cytosine forms hydrogen bonds with guanine. Due to specific pairing of bases the two strands are complementary.

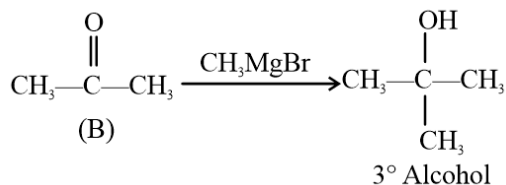
Q94 Text Solution:

(3)



CAN test for alcohol : ✓

Iodoform test : ✓



CAN test for alcohol : ✓

Lucas test : Immediately

Iodoform test : x

Q95 Text Solution:

(1)

For acidic buffer, $pH = pK_a + \log \frac{[Salt]}{[Acid]}$ As $[Salt] = [Acid]$

$$\therefore pH = pK_a = 9.30$$

Q96 Text Solution:

(1)

(i) Lucas test $\rightarrow$ conc. $HCl + ZnCl_2$

(ii) Dumas method $\rightarrow CuO/CO_2$

(iii) Kjeldahl's method $\rightarrow H_2SO_4$

(iv) Hinsberg Test $\rightarrow C_6H_5SO_2Cl + KOH$

Q97 Text Solution:

(2)

$$A(g) \rightleftharpoons B(g) + \frac{1}{2}C(g)$$

$$t = t_{eq} \cdot \frac{1-\alpha}{\alpha} \cdot \frac{\alpha}{\frac{1}{2}}$$

$$P_B = \frac{\alpha}{\left(1 + \frac{\alpha}{2}\right)} \cdot P,$$

$$P_A = \frac{(1-\alpha)}{\left(1 + \frac{\alpha}{2}\right)} \cdot P,$$

$$P_C = \frac{\frac{\alpha}{2}}{\left(1 + \frac{\alpha}{2}\right)} \cdot P$$

$$K_P = \frac{P_B \cdot P_C^{\frac{1}{2}}}{P_A} = \frac{\left(\alpha\right)^{\frac{3}{2}} \left(P\right)^{\frac{1}{2}}}{\left(1-\alpha\right) \left(2+\alpha\right)^{\frac{1}{2}}}$$

Q98 Text Solution:

(4)

i-(D), ii-(C), iii-(B), iv-(A)

Q99 Text Solution:

(3)

$$\text{Energy of one photon} = \frac{hc}{\lambda} = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{5893 \times 10^{-10}} J$$

$$50 \text{ Watt} = 50 \text{ Js}^{-1}$$

Energy emitted by sodium lamp in one second



$$= 50 \text{ J}$$

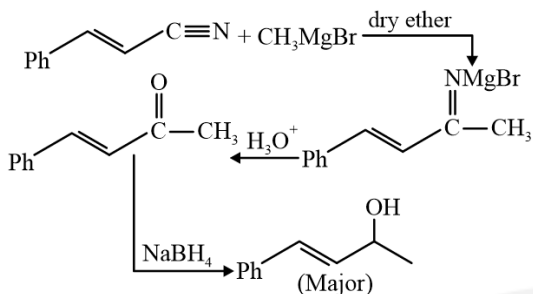
$$\text{Energy emitted in 5 hours} = 50 \times 5 \times 60 \times 60 \text{ J}$$

Number of photons emitted in 5 hours

$$= \frac{\text{Total energy emitted}}{\text{Energy of one photon}} = \frac{50 \times 5 \times 3600 \times 5893 \times 10^{-10}}{6.6 \times 10^{-34} \times 3 \times 10^8} = 2.678 \times 10^{24} \text{ photons}$$

Q100 Text Solution:

(3)



Q101 Text Solution:

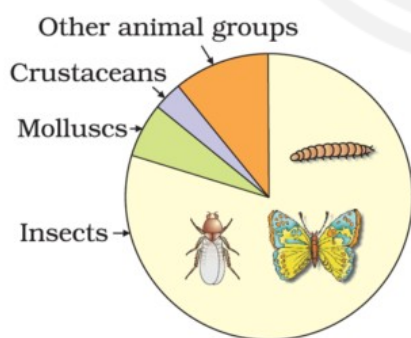
(4)

Some examples of recent extinctions include the dodo (Mauritius), quagga (Africa), thylacine (Australia), Steller's Sea Cow (Russia) and three subspecies (Bali, Javan, Caspian) of tiger. Many species extinctions in the last 500 years (Steller's sea cow, passenger pigeon) were due to overexploitation by humans.

[New NCERT Class 12th Page No. 221,223]

Q102 Text Solution:

(3)



[New NCERT Class 12th Page No. 218]

Q103 Text Solution:

(4)

Interaction of biotic and abiotic components is characteristic for each type of ecosystem. Identification and enumeration of species of an ecosystem gives its species composition.

[New NCERT Class 12th Page No. 206]

Q104 Text Solution:

(1)

Francis Crick postulated the presence of an adapter molecule that would on one hand read the code and on other hand would bind to specific amino acids.

[New NCERT Class 12th Page No. 98]

Q105 Text Solution:

(3)

If detritus is rich in lignin and chitin decomposition rate is slow.

Warm and moist environment favour decomposition whereas low temperature and anaerobiosis inhibit decomposition resulting in build up of organic materials.

[New NCERT Class 12th Page No. 208]

Q106 Text Solution:

(2)

Cuckoo and crow show brood parasitism.

[New NCERT Class 12th Page No. 200]

Q107 Text Solution:

(1)

At the ecosystem level, India, for instance, with its deserts, rain forests, mangroves, coral reefs, wetlands, estuaries, and alpine meadows has a greater ecosystem diversity than a Scandinavian country like Norway.

[New NCERT Class 12th Page No. 217]

Q108 Text Solution:

(4)

Endemism	Species found in a particular area only
Hotspot of India	Western Ghats
Sacred groove	Khasi and Jaintia hills Meghalaya
Cryopreservation	Advanced <i>ex-situ</i> conservation

[New NCERT Class 12th Page No. 224,225]

Q109 Text Solution:

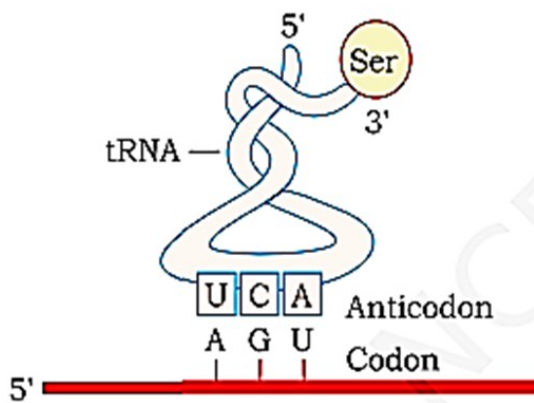
(2)



Android App

iOS App

PW Website



In charged tRNA amino acid is present at 3' position.

[New NCERT Class 12th Page No. 98]

Q110 Text Solution:

(3)

Primary consumers are herbivores.

[New NCERT Class 12th Page No. 210]

Q111 Text Solution:

(4)

The vertical distribution of different species occupying different levels is called stratification.

[New NCERT 12th Page No. 206]

Q112 Text Solution:

(3)

Some species of insects and frogs are cryptically-coloured (camouflaged) to avoid being detected easily by the predator.

[New NCERT Class 12th Page No. 198]

Q113 Text Solution:

(2)

Total number of gene(estimated)	30,000 genes
Average gene consists of	3000 bases
Chromosome 1	2968 genes
Chromosome Y	231 genes

[New NCERT Class 12th Page No. 104]

Q114 Text Solution:

(2)

Decomposers secrete digestive enzymes that breakdown dead and waste materials into simple, inorganic materials, which are subsequently absorbed by them.

[New NCERT Class 12th Page No. 210]

Q115 Text Solution:

(4)

Heterochromatin are densely packed and stain dark.

[New NCERT Class 12th Page No. 84]

Q116 Text Solution:

(2)

In an aquatic ecosystem, GFC (grazing food chain) is the major conduit for energy flow.

[New NCERT Class 12th Page No. 210]

Q117 Text Solution:

(4)

There is single DNA-dependent RNA polymerase that catalyses transcription of all types of RNA in bacteria.

[New NCERT Class 12th Page No. 93]

Q118 Text Solution:

(2)

The ecosystem varies greatly in size from a small pond to a large forest or a sea.

[New NCERT Class 12th Page No. 205]

Q119 Text Solution:

(3)

A nucleotide has a nitrogenous base, phosphate group, a ribose sugar.

[New NCERT Class 12th Page No. 80]

Q120 Text Solution:

(2)

It is showing declining population age pyramid.

[New NCERT Class 12th Page No. 192]

Q121 Text Solution:

(4)

The amount of energy decreases at successive trophic levels.

[New NCERT Class 12th Page No. 211]

Q122 Text Solution:

(4)

India now has 14 biosphere reserves, 90 national parks and 448 wildlife sanctuaries.

[New NCERT 12th Page No. 225]

Q123 Text Solution:

(4)

AUG	Methionine
UAA	Stop codon
UUU	Phenylalanine
UAC	Tyrosine

[New NCERT Class 12th Page No. 96]

Q124 Text Solution:



(1)

The annual net primary productivity of the whole biosphere is approximately 170 billion tons (dry weight) of organic matter.

[New NCERT 12th Page No. 207]

Q125 Text Solution:

(2)

Gause's 'Competitive Exclusion Principle' states that two closely related species competing for the same resources cannot co-exist indefinitely and the competitively inferior one will be eliminated eventually.

[New NCERT Class 11th Page No. 199]

Q126 Text Solution:

(3)

Western Ghat regions of Karnataka and Maharashtra

The Sarguja, Chanda and Bastar areas of Madhya Pradesh.

[New NCERT Class 12th Page No. 225]

Q127 Text Solution:

(2)

Emigration is the number of individuals of the population who left the habitat and gone elsewhere during the time period under consideration.

[New NCERT Class 12th Page No. 193]

Q128 Text Solution:

(2)

Thymine is present in DNA only and uracil is present in RNA only.

[New NCERT Class 12th Page No. 81]

Q129 Text Solution:

(3)

$$dN / dt = rN \left(\frac{K-N}{K} \right).$$

[New NCERT 12th Page No. 195]

Q130 Text Solution:

(1)

DNA are separated from bulk genomic DNA as different peaks during density gradient centrifugation. The bulk DNA forms a major peak and the other small peaks are referred to as satellite DNA.

[New NCERT Class 12th Page No. 105]

Q131 Text Solution:

(1)

A considerable amount of GPP is utilised by plants in respiration. Gross primary productivity minus respiration losses (R), is the net primary productivity (NPP).

$$GPP - R = NPP$$

[New NCERT Class 12th Page No. 207]

Q132 Text Solution:

(3)

The Mediterranean orchid *Ophrys* employs 'sexual deceit' to get pollination done by a species of bee. One petal of its flower bears an uncanny resemblance to the female of the bee in size, colour and markings. The male bee is attracted to what it perceives as a female, 'pseudocopulates' with the flower, and during that process is dusted with pollen from the flower. When this same bee 'pseudocopulates' with another flower, it transfers pollen to it and thus, pollinates the flower.

[New NCERT Class 12th Page No. 202]

Q133 Text Solution:

(4)

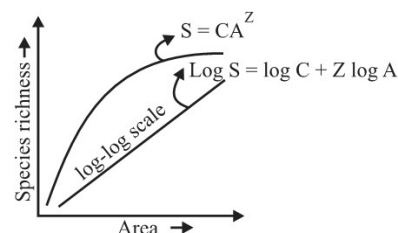
Of the incident solar radiation less than 50 per cent of it is photosynthetically active radiation (PAR).

Except for the deep sea hydro-thermal ecosystem, sun is the only source of energy for all ecosystems on Earth.

[New NCERT Class 12th Page No. 209]

Q134 Text Solution:

(4)



[New NCERT Class 12th Page No. 220]

Q135 Text Solution:

(3)

In the Alfred Hershey and Martha Chase experiments the following isotopes were used ³⁵S, ³²P.

[New NCERT Class 12th Page No. 86]

Q136 Text Solution:

(1)

Mango	1,000 varieties in India
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Rice	50,000 varieties in India
International Union for Conservation of Nature and Natural Resources	More than 1.5 million species described (plants and animals)
Global species diversity	7 million

[New NCERT Class 12th Page No. 217]

Q137 Text Solution:

(2)

The *mycorrhizae* are associations between fungi and the roots of higher plants.

[New NCERT Class 12th Page No. 201]

Q138 Text Solution:

(3)

The historic Convention on Biological Diversity ('The Earth Summit') was held in Rio de Janeiro. The World Summit on Sustainable Development was held in 2002 in Johannesburg, South Africa.

[New NCERT Class 12th Page No. 225]

Q139 Text Solution:

(2)

A wide variety of chemical substances that we extract from plants on a commercial scale (nicotine, caffeine, quinine, strychnine, opium, etc.,) are produced by them actually as defences against grazers and browsers.

[New NCERT Class 12th Page No. 198]

Q140 Text Solution:

(2)

In a field experiment, when all the starfish were removed from an enclosed intertidal area, more than ten species of invertebrates became extinct within a year, because of interspecific competition.

[New NCERT Class 12th Page No. 198]

Q141 Text Solution:

(1)

By the process of leaching, watersoluble inorganic nutrients go down into the soil horizon and get precipitated as unavailable salts. Plants capture only 2-10 per cent of the PAR. Unidirectional flow of energy from the sun to producers and then to consumers.

[New NCERT Class 12th Page No. 207, 209]

Q142 Text Solution:

(2)

Connell's elegant field experiments showed that on the rocky sea coasts of Scotland, the larger and competitively superior barnacle *Balanus* dominates the intertidal area, and excludes the smaller barnacle *Chthamalus* from that zone.

[New NCERT Class 12th Page No. 199]

Q143 Text Solution:

(2)

Once covering more than 14 per cent of the earth's land surface, these rain forests now cover no more than 6 per cent.

[New NCERT Class 12th Page No. 222]

Q144 Text Solution:

(1)

The Monarch butterfly is highly distasteful to its predator (bird) because of a special chemical present in its body. Interestingly, the butterfly acquires this chemical during its caterpillar stage by feeding on a poisonous weed.

[New NCERT Class 12th Page No. 198]

Q145 Text Solution:

(4)

Bioprospecting is exploring molecular, genetic and species-level diversity for products of economic importance.

[New NCERT Class 12th Page No. 223]

Q146 Text Solution:

(2)

The tRNA binds to specific amino acid at one end and pairs through H-bonding with codes on mRNA through its anticodons.

[New NCERT Class 12th Page No. 108]

Q147 Text Solution:

(4)

In general, loss of biodiversity in a region may lead to:

(a) decline in plant production, (b) lowered resistance to environmental perturbations such as drought and (c) increased variability in certain ecosystem processes such as plant productivity, water use, and pest and disease cycles.

[New NCERT Class 12th Page No. 222]

Q148 Text Solution:

(1)

The lac operon consists of one regulatory gene (the *i* gene – here the term *i* does not refer to inducer, rather it is derived from the word inhibitor).



[New NCERT Class 12th Page No. 100]

Q149 Text Solution:

(2)

Obtaining industrial products like tannins, lubricants, dyes, resins, perfumes come under narrowly utilitarian argument.

[New NCERT Class 12th Page No. 223]

Q150 Text Solution:

(2)

The DNA in nucleoid is organised in large loops held by proteins.

[New NCERT Class 12th Page No. 83]

Q151 Text Solution:

(3)

The appearance of dry, scaly lesions on various body parts, is seen in ringworm infections.

[New NCERT Class 12th Page no. 133]

Q152 Text Solution:

(4)

- The chemical carcinogen present in tobacco smoke is identified as a major cause of lung cancer.
- Cellular oncogenes have been identified in normal cells and activated under certain conditions, which could lead to oncogenic transformation of the cell.
- Techniques like CT scans and MRI are beneficial in detecting cancer of the internal organs.
- Antibodies against cancer-specific antigens are also used for the detection of certain cancers.

[New NCERT Class 12th Page No. 141, 142]

Q153 Text Solution:

(3)

- HIV decreases the natural immunity of the body by destroying T-lymphocytes (T_H).
- HIV enters into helper T-lymphocytes (T_H), replicates and produces progeny viruses.
- The progeny viruses released in the blood attack other helper T-lymphocytes. This is repeated, leading to a progressive decrease in the number of helper T-lymphocytes in the body of the infected person.

[New NCERT Class 12th Page No. 140]

Q154 Text Solution:

(1)

Biopiracy is the term used to refer to the use of bio-resources by multinational companies and other organisations without proper authorisation from the countries and people concerned without compensatory payment.

[New NCERT Class 12th Page No.185]

Q155 Text Solution:

(3)

The r-DNA molecule cannot be created, unless one cuts the DNA (source) and vector with same restriction enzyme, because when cut by same enzyme both will have same sticky ends, that can be joined by DNA ligase.

[New NCERT Class 12th Page No. 166]

Q156 Text Solution:

(2)

Typhoid	Pathogens usually enter the small intestine
Pneumonia	Pathogens usually infect alveoli
Common cold	Pathogens infect the nose and respiratory passage only
Amoebiasis	Pathogens usually enter the large intestine

[New NCERT Class 12th Page no. 130, 131, 132]

Q157 Text Solution:

(3)

Primers are small chemically synthesized oligonucleotides and complementary to the template DNA.

[New NCERT Class 12th Page No. 202]

Q158 Text Solution:

(1)

Foreign DNA can be ligated at the *Bam*HI site of the tetracyclin resistance gene in the vector pBR322.

[New NCERT Class 12th Page No. 169]

Q159 Text Solution:

(3)

- Restriction endonucleases are used in genetic engineering to form 'recombinant' molecules of DNA, which are composed of DNA from different sources/genomes.
- DNA is a hydrophilic molecule and it cannot pass through cell membranes.



[New NCERT Class 12th Page No. 174]**Q160 Text Solution:**

(3)

The correct sequence for replication of retrovirus is:

Introduction of viral RNA → production of viral DNA by reverse transcription → incorporation of viral DNA in host genome → new viral RNA produced by the infected cells.

[New NCERT Class 12th Page no. 139]**Q161 Text Solution:**

(3)

Rice is an important food grain, the presence of which goes back thousands of years in Asia's agricultural history. There are an estimated 200,000 varieties of rice in India alone. The diversity of rice in India is one of the richest in the world.

[New NCERT Class 12th Page No. 185]**Q162 Text Solution:**

(1)

- Morphine is commonly used as a painkiller after surgery due to its analgesic properties.
- Chewing tobacco typically increases blood pressure and heart rate due to the nicotine it contains.
- Cocaine is not commonly given post-surgery; it can have harmful effects and is not used as a stimulant for recovery.
- Barbiturates can be used to treat insomnia.

[New NCERT Class 12th Page No. 143, 144]**Q163 Text Solution:**

(1)

Excessive alcohol consumption may lead to liver cirrhosis.

[New NCERT Class 12th Page no. 146]**Q164 Text Solution:**

(3)

- DNA is a negatively charged molecule, so they can be separated by forcing them to move towards the anode under an electric field.
- DNA fragments separate according to size through the pores of agarose gel.
- The separated DNA fragments can be visualized only after staining the DNA with a compound known as ethidium bromide then followed by exposure to UV radiation.

[New NCERT Class 12th Page No. 198]**Q165 Text Solution:**

(3)

- The heat shock method is used in bacterial transformation to create temporary pores in the bacterial cell membrane, allowing the uptake of foreign DNA, such as plasmids.
- The sudden increase in temperature disrupts the membrane briefly, enabling the DNA to enter the bacterial cell.

[New NCERT Class 12th Page No. 171]**Q166 Text Solution:**

(1)

An alien DNA replicate and multiply within the host organism by integrating into the host's chromosome that has an origin of replication.

[New NCERT Class 12th Page No. 164]**Q167 Text Solution:**

(3)

- Bt cotton contains the *cry* gene from *Bacillus thuringiensis*, which produces a protein toxic to insects like cotton bollworms.
- The insecticidal protein kills insects by disrupting their gut cells.

[New NCERT Class 12th Page No. 180]**Q168 Text Solution:**

(3)

- The proteins encoded by the genes *cryIAC* and *cryIIAb* control the cotton bollworms, that of *cryIAb* controls corn borer.

[New NCERT Class 12th Page No. 178]**Q169 Text Solution:**

(3)

- Recombinant DNA is inserted within the coding sequence of an enzyme, β -galactosidase and the presence of insert results into insertional inactivation of the β -galactosidase gene and the colonies do not produce any colour in the presence of chromogenic substrate.
- At present, about 30 recombinant therapeutics have been approved for human-use the world over.

[New NCERT Class 12th Page No. 181, 183]**Q170 Text Solution:**

(3)

- Decreased production in males is a specific side effect in males, as females do not produce sperm.
- The other side effects, such as increased aggressiveness, mood swings, and depression, can occur in both males and females using anabolic steroids.

[New NCERT Class 12th Page no. 146]**Q171 Text Solution:**

(3)

Cancer cells typically exhibit uncontrolled growth, can invade surrounding tissues, and often form tumors.

[New NCERT Class 12th Page no. 141]**Q172 Text Solution:**

(1)

The palindromes in DNA are base pair sequences that are same when read forward (left to right) or backward (right to left) from a central axis of symmetry.

5' - GAATTC - 3'

3' - CTTAAG - 5'

[New NCERT Class 12th Page No. 195]**Q173 Text Solution:**

(3)

- Plague is a bacterial disease.
- Rheumatoid arthritis is an auto-immune disease.

[New NCERT Class 12th Page no. 131, 137]**Q174 Text Solution:**

(1)

If we ligate a foreign DNA at *Pvu*I site of pBR322, the recombinant plasmid will lose resistance for ampicillin.

[New NCERT Class 12th Page No. 199]**Q175 Text Solution:**

(1)

- Polymerases synthesises DNA or RNA strands, not break DNA into fragments.
- Nucleases degrades nucleic acids by breaking phosphodiester bonds, but they don't separate the two strands of DNA.
- Lysozyme breaks down the bacterial cell wall.

[New NCERT Class 12th Page No. 195]**Q176 Text Solution:**

(2)

The common cold is a viral disease, caused by a group of viruses called rhinoviruses.

[New NCERT Class 12th Page no. 131]**Q177 Text Solution:**

(3)

- RNA interference takes place in all eukaryotic organisms and that involves silencing of specific mRNA.
- In mammals, insulin is synthesised as a pro-hormone that needs to be processed before it becomes a fully mature and functional hormone.

[New NCERT Class 12th Page No.180, 181]**Q178 Text Solution:**

(2)

Genetic Engineering Approval Committee is responsible for overseeing the safety of genetically modified organisms and ensuring that genetic engineering practices are safe for human health and the environment.

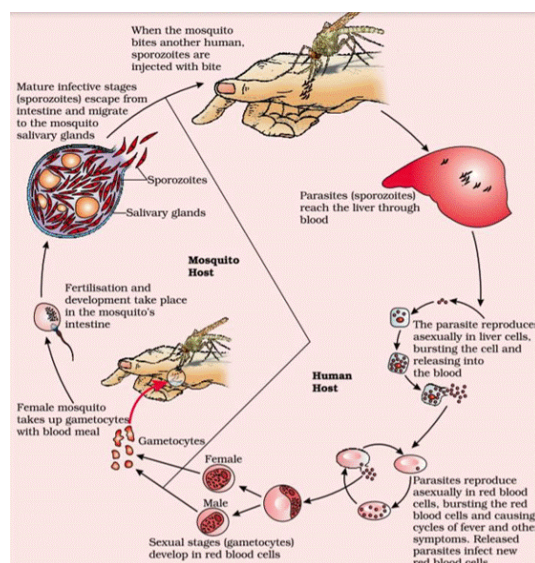
[New NCERT Class 12th Page No. 184]**Q179 Text Solution:**

(3)

Smoking causes a deficiency of oxygen in the body because it increases the carbon monoxide (CO) content in the blood and reduces the concentration of haembound oxygen.

[New NCERT Class 12th Page no. 144]**Q180 Text Solution:**

(4)



The asexual reproduction takes place in both liver and RBCs.

[New NCERT Class 12th Page no. 132]

Q181 Text Solution:

(4)

- Peyer's patches are secondary lymphoid organs.
- The bone marrow and thymus are involved in the development and maturation of lymphocytes.
- IgE antibodies are associated with allergic responses.
- The receptors for cannabinoids are principally present in the brain.
- The drugs can be obtained from fungi.

[New NCERT Class 12th Page no. 137, 142, 143]

Q182 Text Solution:

(3)

- The antibodies produced in response to vaccination provide long-term immunity against the antigens that would neutralise the pathogenic agents during actual infection.
- The vaccines generate memory B and T -cells that recognise the pathogen quickly on subsequent exposure and overwhelm the invaders with a massive production of antibodies.

[New NCERT Class 12th Page no. 136]

Q183 Text Solution:

(3)

- Nematode-specific genes were introduced into the host plant by using *Agrobacterium*.
- A single stranded DNA or RNA, tagged with a radioactive molecule (probe) is allowed to hybridise to its complementary DNA in a clone of cells followed by detection using autoradiography.

[New NCERT Class 12th Page No. 180]

Q184 Text Solution:

(2)

- Traditional knowledge enhances biotechnological advancements and reduces research and development costs in agriculture and medicine.
- It primarily benefits developing countries and promotes biodiversity, making it valuable for

scientific research.

[New NCERT Class 12th Page No. 184 -186]

Q185 Text Solution:

(3)

Loss of the property of the body to recognise and differentiate between self-cells and non-self cells may lead to autoimmune diseases.

[New NCERT Class 12th Page no. 137]

Q186 Text Solution:

(3)

- The recombinant DNA vector carries a gene for ampicillin resistance.
- When the *E. coli* cells are transformed with this vector and then spread on agar plates containing ampicillin, only those cells that have taken up the recombinant DNA and are thus resistant to ampicillin will survive and grow.
- Non-transformed cells, which do not have the ampicillin resistance gene, will be killed by the ampicillin and will not grow.

[New NCERT Class 12th Page No. 165]

Q187 Text Solution:

(2)

Cocaine interferes with the transport of the neurotransmitter dopamine.

[New NCERT Class 12th Page no. 143]

Q188 Text Solution:

(1)

In the plasmid pBR322, the *rop* gene codes for proteins involved in regulating the replication of the plasmid.

[New NCERT Class 12th Page No. 199]

Q189 Text Solution:

(2)

- Downstream processing and quality control vary depending on the product, as mentioned in the original text.
- As different products require different processes based on their nature and use.

[New NCERT Class 12th Page No. 204, 205]

Q190 Text Solution:

(2)

Mucosa-associated lymphoid tissue (MALT) constitutes about 50 per cent of lymphoid tissue in the human body.



[New NCERT 12th Page no. 138]**Q191 Text Solution:**

(2)

The Indian Parliament has recently cleared the second amendment of the Indian Patents Bill, that takes such issues into consideration, including patent terms emergency provisions and research and development initiative.

[New NCERT Class 12th Page No. 185]**Q192 Text Solution:**

(3)

- Metastasis is the most feared property of malignant tumours.
- In metastasis, cells sloughed from malignant tumours reach distant sites through blood, and spread a new tumour at other sites of the body.

[New NCERT Class 12th Page no. 141]**Q193 Text Solution:**

(3)

The main objective of inserting alien DNA into a cloning vector and transferring it into a host cell is to express and produce a desirable protein. This involves creating recombinant DNA that directs the host cell to make a specific protein.

[New NCERT Class 12th Page No. 173]**Q194 Text Solution:**

(1)

Emphysema	Can be treated by α -1 antitrypsin
Rosie	Protein enriched milk
ELISA	Based on antigen-antibody interaction.
PCR	Amplification of DNA.

[New NCERT Class 12th Page No. 181, 184]**Q195 Text Solution:**

(2)

Sustained high fever (39° to 40°C) weakness, stomach pain, constipation, headache, loss of appetite and intestinal perforations in severe cases are symptoms of typhoid caused by the bacterium *Salmonella typhi*.

[New NCERT Class 12th Page no. 130]**Q196 Text Solution:**

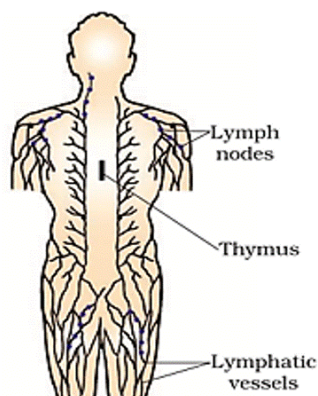
(1)

Each restriction endonuclease recognises a specific palindromic nucleotide sequences in the

DNA.

[New NCERT Class 12th Page No. 165]**Q197 Text Solution:**

(4)



The thymus is a lobed organ located near the heart and beneath the breastbone and quite large at the time of birth but keeps reducing in size with age.

[New NCERT Class 12th Page No. 138]**Q198 Text Solution:**

(3)

- The foetus receives some antibodies from their mother, through the placenta during pregnancy and provides passive immunity to the foetus.
- Active immunity is slow and takes time to give its full effective response.

[New NCERT Class 12th Page no. 136]**Q199 Text Solution:**

(1)

Physiological barrier	Tears from eyes
Physical barrier	Skin
Cellular barrier	Macrophages in tissues
Cytokine barrier	Interferons

[New NCERT Class 12th Page no. 134, 135]**Q200 Text Solution:**

(1)

- Immunity is the overall ability of the host to fight the disease-causing organisms conferred by the immune system.
- The primary immune response is of low intensity.

[New NCERT Class 12th Page no. 131, 134, 135][Android App](#)[iOS App](#)[PW Website](#)



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